FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000176
Award Program	Seeding Solutions – Urban Agriculture
Project Title	A community-based microfarm aggregation model for transforming agricultural production and enhancing asset based economic development in post-industrial urban areas
Grant Start Date	01/01/2019
Grant End Date	12/31/2022
Total Grant Budget	$ 2,068,336.11
Total FFAR Budget	$ 962,600
Matching Funder(s)	
Final Report Date	04/03/2023
Project Director/Principal Investigator Information	
Full Name	Kent Curtis
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Grantee Organization Information	
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Authorized Signing Official (ASO) Information	
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ASO Title	Assistant Director-Office of Research Office of Sponsored Programs
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General Information

1.1. Please list the geographic location(s) – city, state, congressional district - wherethe work was conducted. If the work was conducted outside of the US, please list the city and country.

Work Conducted at The Ohio State University-Regional Mansfield Campus in Mansfield, OH which is US Congressional District OH-012.
The Ohio State University applicant is located in US Congressional District OH-003

1.2. How many new jobs were created because of this grant funding?

12

1.3. How many jobs were maintained because of this grant funding?

12

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profitstatus since you were awarded the grant? If yes, please explain.

No

1.5. Has your organization undergone a recent name change? If so, please providethe new name of your organization.

No

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

The project proposed to build and measure the impacts of a multi-microfarm aggregation system (cooperative) in a post-industrial small city. We succeeded in building a sustainable system, converting a 12-acre brownfield into a vibrant urban farm, and training and engaging six beginning farmers to produce commercially in the microfarm setting and participate productively in a crop aggregation planning model. The cooperative is now poised to expand through the addition of new rural growers and the training and onboarding of new microfarmers. While it was underestimated significantly in our proposal (and the COVID-19 interruptions wreaked havoc on our schedule) the process of training new farmers, installing microfarms, developing a marketing strategy that brought the most sustainable price points, developing protocols for cooperative operations, developing tools to help the whole system function efficiently, and having all of the pieces sufficiently in place to insure buyer relationships took all three years and is continuing to the present day with additional local support.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Our key findings through this process relate to four areas:

1. Developed and refined a training curriculum to identify and prepare beginning farmers to operate a commercial microfarm business as a member-owner of a farmer cooperative.
2. Created standard operating protocols for farmer-members and for cooperative data tracking and management.
3. Calculated development costs and strategy for system replication.
4. Ground verified production data from 2020, 2021, and 2022 and recalculated the production potential for a 6,000-8,000 square foot microfarm.

Another key finding was the critical need for a cooperative-wide crop data management tool. During our preliminary trainings, we trained farmers on the farm with different data collection protocols than in our cooperative workshops – an oversight that went unnoticed until late in 2020 when a review of annual operations on the individual farms were not always consistent with the production data entered at the aggregation site. To remedy this initially, the aggregation manager was tasked with taking the bulk orders for the year ahead and developing a crop plan across all the farms, which was then distributed to each farm. There was some confusion in some locations about how to use this data and in 2021, the process was simplified to indicate to the farmer their precise planting dates for each crop and how many square feet needed to be cultivated to produce the needed harvests.

This system is working effectively but demands an enormous amount of cooperative personnel time during the winter months and continues to encounter occasional hiccups in action. We have recently created a more simplified tool, using excel, that provides the farm with the ability to plan out their entire farm, bed by bed, season by season, based on taking a share of the bulk orders at the beginning of the year. We are training Marion farmers (our second post-FFAR microfarm implementation site) to use this tool and we have begun training the Richland Gro-Op to use it as well. We believe a tool that can simplify crop planning across the cooperative while placing ownership for individual farm plans on the farmers themselves puts the appropriate knowledge in the appropriate places.

While this pilot effort required almost $2 million to launch over three years, based on the experiences and knowledge gained during this process, we now estimate that a small city could launch a similar system with the training program we developed and get similar results for approximately $500,000, with local financing and asset sharing. To prove this concept, we have worked with the small city of Marion, also home to an Ohio State regional campus, to launch a similar system. Farmers are starting their second year of training in 2023 and we have raised $200,000 so far toward the effort.

When we estimated the production potential for a standard microfarm, we based it on the 3,000 square foot demonstration microfarm built on the Ohio State Mansfield campus in 2017. As implementation and production proceeded, we developed larger beds and closed the space between beds in our farms, netting another 3,000 to 5,000 square feet out of approximately the same footprint.

In 2017, we estimated that our demonstration farm might be capable of producing a $35,000/yr net income for a farmer. Based on two years of farm production and harvest data and two years of price-point data, this enlarged space is now calculated to have the potential to produce closer to $40,000 net income a year for cooperative members. Based on our experiences with growers in Mansfield and Marion, this income is dependent on strong buyer relationships in several sectors – home produce delivery, high end restaurant, catering produce distributors, direct marketing, value added commodities – and depends upon the farmer meeting the harvest goals for every crop across the entire season. The Richland Gro-Op has secured the first part and its farmers have pared themselves down to their best producers and have created incentives to bring on new growers to assure the second part during 2023.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that couldimpact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices,legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

We have refined a comprehensive and field-tested beginning farmer training program around the microfarm and cooperative system that can be replicated in small and medium-sized cities around the United States and world either through engaged food system not for profit organizations or by Land Grant Extension educators.

We have developed a clear understanding of the costs involvedd for implementing such a project based on two field implementation efforts (Mansfield and Marion) and have refined productivity estimates for microfarms and microfarming system based on production data from 2019-2022.

The microfarm system has the potential to re-invigorate small farm and urban agriculture in small and medium sized cities in the United States by offering a model cooperative approach, sufficient and effective beginning farmer training, and a community engaged approach that embeds the work in communities and secures its long-term sustainability through rigorous market participation.

The microfarm site and the aggregation model also present a rich arena of digital research. Currently microfarms are estimated to have a potential to produce more than $40,000 a year in supplemental income. With a strategically designed data-streaming and automation system – smart-beds, automated watering a nutrients, a modular automous farm robot – this income potential could be increased significantly by freeing up time from repetitive labor. We are currently exploring these potentials as well as investigating the integration of hydroponics into the microfarm design.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

<u>Production System Goals</u>
Year 1 –
Obj. 1: Assemble the producers and components of production model for establishing a sustainable microfarm system.
Obj 2: Coordinate production of one fall crop 2019.

a. Order and assemble microfarm kits with producers
b. Recruit two additional producers (12 already recruited)
c. Offer educational workshops with Ohio Extension Educators on high tunnel management, crop selection for economic return and labor evenness, organics recycling, fertility management, integrated pest management
d. Formalize local farmer-owned co-operative organization and membership
e. Hire marketing agent
f. Hire cooperative coordinator to oversee marketing, communication, farmer coordination
g. Plan and plant coordinated fall crop
h. Evaluate growing and aggregation system based on fall production
i. Measure and evaluate organics recycling potential in system
j. Plan Year 2 full year production

Year 2 –

Obj. 1: Test a model a replicable model for establishing a sustainable microfarm.
Obj. 2: Coordinate production for the full year 2020.

a. Develop an integrated organics recycling system with microfarmers, local food producers, municipalities, and education institutions to support microfarm soils and nutrient needs
b. Complete full year (2020) of production and marketing

Year 3 –

Obj. 1: Optimize a model a replicable model for establishing a sustainable microfarm
Obj. 2: E*valuate*, *model* and *coordinate production for the full year 2021 adjusted based on lessons from 2020.*

<u>Research Projects Goals</u>

Year 1 – *To gather base data to enhance our participatory community* *development approach that emphasizes identifying strengths that exist in the community, opportunities*

Obj. 1: Develop a detailed understanding of spatial areas and issues inhibiting healthy food access in Mansfield.

a. FEAST (Food-mapping for Empowerment, Access, and Sustainable Transformation) baseline data collection and mapping to understand healthy food access across Mansfield communities using participatory and mixed-methods approach to begin identifying ways to address the lack of access.

Obj. 2: Create a collaborative and democratic community engagement process that includes community household participation.

a. Household Resiliency Survey I will expand on the FEAST participatory action with input from community households to identify strengths, opportunities, and challenges faced by cross the community.
b. Community meetings access/healthy choices to initiate engagements with community members and groups.

Obj. 3: Identifying, engaging, and harnessing farming expertise and resources of Richland County and surrounding agricultural communities.

a. Urban farmer interviews and focus groups will be conducted to assess information, expertise, financial, and technology needs of new urban farmers through one-on-one interviews then later in focus groups.
b. Rural farmer interviews and focus groups will be conducted to identify community resources for agriculture that can meet the needs of urban farmers. Identify synergisms between urban and rural farmers to make urban agriculture development complementary to rural farmers and not competitive.
c. Rural/Urban farmer meetings to assess perceptions and experiences from the first year and facilitate input to shape the second year.

Obj. 4: Collect biophysical data to develop and support urban microfarming production systems.

a. Landscape remote imaging data baseline collection and Geographic Information System development for the project to provide quantitative analysis and indexing of land cover, climate, and soil suitability for specific urban farming enterprises.
b. Examine issues for handling microfarm waste (i.e. weeds, non-food parts of plants, food waste) for composting organic materials to harness the nutrients as fertilizer for future microfarm use.

Year 2 – *To gather second year data and begin data analysis*

Obj. 1: Use of detailed understanding of spatial areas and issues inhibiting healthy food access in Mansfield.

a. Develop and utilize the FEAST (Food-mapping for Empowerment, Access, and Sustainable Transformation) mapping system to enhance participation of Mansfield community members and assist collaborations with community members to address healthy food access across Mansfield.

Obj. 2: Implement and expand the collaborative and democratic community engagement process that includes community household participation:

a. Household Resiliency Survey II will expand on the prior year survey and enhance the FEAST mapping system with input from community households to identify strengths, opportunities, and challenges faced by cross the community.
b. Community meetings access/healthy choices to continue engagements with community members and groups.

Obj. 3: Engaging, harnessing, and facilitating the integration of farming expertise and resources of Richland County and surrounding agricultural communities.

a. Objective Rural/Urban farmer meetings to bring together the experiences and developing expertise of urban farmers with existing expertise of rural farmers as information sources. This will be done to build on existing interest among rural farmers to participate in a mentorship program.
b. Rural/Urban farmer meetings to assess perceptions and experiences from the first year and facilitate input to shape the second year.

Obj. 4: Evaluate data for use in supporting urban microfarming production systems.

a. Landscape remote imaging data baseline collection and Geographic Information System development for the project to provide quantitative analysis and indexing of land cover, climate, and soil suitability for specific urban farming enterprises.
b. Address issues for handling microfarm waste (i.e. weeds, non-food parts of plants, food waste) and develop systems for composting organic materials to harness the nutrients as fertilizer for future microfarm use.

Year 3 – To gather third year data and begin data analysis to enhance and evaluate the programs

Obj. 1: Continued use of detailed understanding of spatial areas and issues inhibiting healthy food access in Mansfield.

a. Enhance and utilize the FEAST (Food-mapping for Empowerment, Access, and Sustainable Transformation) mapping system for participation of Mansfield community members.

Obj. 2: Continue expanding the community engagement process.

a. Community meetings access/healthy choices to continue engagements with community members and groups.

Obj. 3: Evaluating the integration of farming expertise and resources of Richland County and surrounding agricultural communities.

a. Host rural/urban farmer meetings to continue facilitating the sharing of experiences and expertise of rural and urban farmers. Continued facilitation of new farmer mentorship program.
c. Host rural/urban farmer meetings to assess perceptions and experiences from the second year and facilitate input to shape the third year.
d. Brief interviews with farmers to evaluate the successes and challenges experienced by each group in participating in various aspects of the project.

Obj. 4: Evaluate and revise information systems developed to support urban microfarming production systems.
b. Landscape remote sensing work for Year II. Analysis to continue providing quantitative analysis for specific urban farming enterprises.
c. Evaluate successes and challenges for handling microfarm waste and the developed systems for composting from farmers and researchers.

Community Engagement Goals

Year 1, 2 & 3 – To keep key stakeholders in community apprised of project development

Obj. 1: Annual Project Summit

2.5. Have any of the major goals/specific aims or milestones for the project changedsince the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Obj. 1: Develop a detailed understanding of spatial areas and issues inhibiting healthy food access in Mansfield.
b. FEAST (Food-mapping for Empowerment, Access, and Sustainable Transformation) baseline data collection and mapping to understand healthy food access across Mansfield communities using participatory and mixed-methods approach to begin identifying ways to address the lack of access.
Obj. 2: Create a collaborative and democratic community engagement process that includes community household participation.
c. Household Resiliency Survey I will expand on the FEAST participatory action with input from community households to identify strengths, opportunities, and challenges faced by cross the community.
d. Community meetings access/healthy choices to initiate engagements with community members and groups.
Obj. 3: Identifying, engaging, and harnessing farming expertise and resources of Richland County and surrounding agricultural communities.

d. Urban farmer interviews and focus groups will be conducted to assess information, expertise, financial, and technology needs of new urban farmers through one-on-one interviews then later in focus groups.
e. Rural farmer interviews and focus groups will be conducted to identify community resources for agriculture that can meet the needs of urban farmers. Identify synergisms between urban and rural farmers to make urban agriculture development complementary to rural farmers and not competitive.
f. Rural/Urban farmer meetings to assess perceptions and experiences from the first year and facilitate input to shape the second year.

Obj. 4: Collect biophysical data to develop and support urban microfarming production systems.

c. Landscape remote imaging data baseline collection and Geographic Information System development for the project to provide quantitative analysis and indexing of land cover, climate, and soil suitability for specific urban farming enterprises.

Most of our social science research goals were significantly impacted by the COVID-19 Pandemic, which shut down our university for several months in 2020 and prevented research activities from beginning until 2021. Preliminary interviews were performed with farmers, but follow-up interviews were not accomplished. The FEAST study had to be shelved altogether, due to its in-person trainings and interview process. In its place, the team developed an online survey gathering data in 2021, but the team has not yet analyzed that data. Remote sensing data was gathered in 2019 and part of 2020, but it also has not been analyzed.

In general, implementation of the farming system and providing the training and support to the group of new farmers who participated in this initiative and to aid them in developing a functioning cooperative consumed much more time and effort than we imagined in the planning process. Federal matching requirements prevented us from installing all ten microfarms during the first year, COVID-19 restrictions restricted hands-on training with the group during 2020, such that only in 2021, the third year of the grant, were all farms installed and farmers trained. For this reason, much of the hoped for social and ecological study of the system itself, which assumed a fully operational cooperative for two years of the grant, is just beginning to become relevant now, as the system has been operational for two years and is beginning to have a sustained (as opposed to a novel) presence in the community. I have spoken with members of the social science team about collecting follow-up data in the year ahead (2023-2024) with which to compare the survey results from 2021.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (bothpositive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Production System Goals

Year 1 –

Obj. 1: Assemble the producers and components of production model for establishing a sustainable microfarm system.

Obj 2: Coordinate production of one fall crop 2019.

a. Order and assemble microfarm kits with producers
b. Recruit two additional producers (12 already recruited)
c. Offer educational workshops with Ohio Extension Educators on high tunnel management, crop selection for economic return and labor evenness, organics recycling, fertility management, integrated pest management
d. Formalize local farmer-owned co-operative organization and membership
e. Hire marketing agent
f. Hire cooperative coordinator to oversee marketing, communication, farmer coordination
g. Plan and plant coordinated fall crop
h. Evaluate growing and aggregation system based on fall production
i. Measure and evaluate organics recycling potential in system
j. Plan Year 2 full year production

Year 2 –

Obj. 1: Test a model a replicable model for establishing a sustainable microfarm.

Obj. 2: Coordinate production for the full year 2020.

a. Develop an integrated organics recycling system with microfarmers, local food producers, municipalities, and education institutions to support microfarm soils and nutrient needs
b. Complete full year (2020) of production and marketing

Year 3 –

Obj. 1: Optimize a model a replicable model for establishing a sustainable microfarm

Obj. 2: E*valuate*, *model* and *coordinate production for the full year 2021 adjusted based on lessons from 2020.*

We accomplished all of the goals outlined in Objective 1 during the first two years of the grant, an adjustment discussed with John Reich during his visit in 2020 and in our previous reports. We trained a dozen beginning farmers to operate their own microfarm. We built a community Urban Farm – farm to school, prison release work, internships, farmer's market (WIC, SNAP), microfarming training, community food waste composting. Our goal was to build all of this in 2019 and measure it during 2020 and 2021, but the complexity of project implementation and the substantial obstacles created by the COVID-19 epidemic together meant that only by 2021 did we have all of the farms operational.

As our beginning farmers launched their operations in 2020 and 2021, we gained a number of key insights about gaps in our farmer training program and mistaken assumptions in our implementation plan. For the farmer training, our heavy focus on commercial horticulture and lighter focus on data management, small business finances, and cooperative membership created a set of circumstances where most of the producers could successfully grow marketable crops, but their volume and timing of harvest were out of sync with expectation at the aggregation site and their financial knowledge about their own operations was undeveloped or absent. During 2021 a complete set of data management and crop management protocols were developed for the whole cooperative and were implemented in 2022, the cooperative's first year operating outside of the FFAR project management, with success. The knowledge gained from this training and implementation process was further enhanced by a Sustainable Agriculture Research and Education Professional Development grant in which we formalized our training curriculum, reviewed and revised its contents based on these experiences, put it through a review and feedback process with beginning farmers in Mansfield and our Extension training team, and revised it. We have been offering this revised curriculum to a group of 15 microfarmers in training in Marion, Ohio since May 2022. Training crops are currently marketed through the Richland Gro-Op and this group anticipates joining RGO as a federated membership in 2024 and launching their own Marion cooperative in 2025 or 2026. PI Kent Curtis anticipates completing a research article outlining the development and implementation of this microfarm curriculum in summer of 2023.

We believe the structure of the project created two persistent difficulties throughout the life of the grant and, as we advance this idea into other similar communities, we have restructured our approach based on these insights. The first decision that led to challenges was our insistence that our group of beginning farmers organize and incorporate their cooperative, the Richland Gro-Op, at the outset of the training process in May 2019. While some of the people knew some of the other people, they were mostly strangers to each other and to us. The initial leadership of the cooperative then took a hostile and antagonistic position toward the project and project management, which took several weeks to resolve, but was ultimately resolved internally by the cooperative board and no further issues arose during the grant. All of the participants agree that significant decisions around cooperative operations and cooperative leadership should have waited a year or two, when everyone involved knew each other much better *and* understood the nature of their farm business engagement more thoroughly. Our beginning farmers in Marion will wait to form a cooperative until their own farm businesses are operating successfully – they can in the meantime market through RGO in Mansfield. The additional data management and farm management training also seeks to better prepare these farmers for cooperative management and participation.

The other decision embedded in the grant that contributed to most of our challenges getting all farms and farmers operating at full capacity was the decision to gift the microfarm kits to the participating trainees at the end of the research period. Our hope was that the value of the investment ($30,000) would provide an incentive for participants to follow our directions and themselves dig in to become as productive as possible. This only turned out to be true for seven of our ten participants. But the performance of the other three – often doing the bare minimum or less in an effort to stay in the program and gain the 'prize' – created problems for the cooperative as a whole. It led to missed produce orders, lost revenue, and bad feelings. Two of the three made their decision to leave the cooperative prior to the completion of the grant and their assets were returned to the project and re-integrated into the existing system. The other farmer only notified the coop this year (2023), after another poor performance last year, that they were withdrawing from the cooperative. It is uncertain what will happen with those assets. A fourth farmers, who did very well as a producer for the cooperative, left the cooperative in 2022 and produced on her own quite successfully in high end farmer's markets and with a few restaurant buyers, but she is growing a portion of her farm in 2023 for RGO again. While giving away the ten microfarm kits has produced seven working farms, the difficulties that just a few poor performers in a cooperative created for the whole cooperative and all of the farmers has led us to develop a different strategic approach in Marion, which we are seeking to build in Mansfield as well. In Marion we are raising funds for a revolving loan fund that farmers who have successfully completed the two-year training process can access. They can finance their microfarm through this fund and can repay their loan straight of the cooperative crop sales. If we had leveraged the funding in the FFAR grant in this way, we would have begun to repay that fund last year.

We allocated a small amount of funding to subsidize the cooperative marketing and coordination during the first two years of the grant as well as cooperative coordination for the life of the grant. While overall production started fairly small in 2019 (6,000 lbs) and more than doubled in 2020 (19,000lbs), due to new farms coming on line. But marketing was a bigger challenge than anticipated after 2019, when the marketing coordinator we had hired left for another job, and our bulk buyer went with her. During 2020 we developed several local relationships, including a high end local restaurant, and we participated in food security programs distributing fresh produce in Columbus, but we had a difficult year with sales nevertheless. We noted that we got our best price point from buyers who wanted our "brand" – local Ohio, community improvement, fresh and naturally grown – and so set out in 2021 to develop those markets. Project management allocated additional personnel funding during 2021 to create market relationships sufficient to maximize the productive capacity of the whole cooperative and this plan was put in place in 2022, the year following the grant. These new market relationships were successful enough in 2022 to create a demand above and beyond the full capacity of RGO farmers in 2023 and the cooperative is now recruiting rural farmers to join the cooperative as producers this year.

In order to understand the operation, we gathered farm-level data on crops and yields and cooperative-level data on aggregation performance (cull rate) and price points. At the end of each year, we reviewed the data with the farmers and discussed outcomes and results. This data helped farmers and project managers see the critical downstream impacts that an early season mistake could make and it also revealed that none of the farms during the three years of the project, ever approached growing and marketing at 100% of their productive capacities. The reasons for the shortages were as different as the farms, but they ranged from failure to prepare ground, to misunderstandings about the crop needs, to failure to prepare plants for frost, to cultivation neglect, to water issues (both too much and too little), to life events, etc. Despite the difficulties getting every part of the system to work in tandem, on the individual farms where the best practices were taking and in pockets of the other farms, application of the cooperative crop plan – that is, planting the crops needed in the amounts needed at the times needed – was netting results. In 2020 and 2021, we aggregated the data from these isolated sites for three dozen crops and used it as a baseline for planning the 2022 season. The goal was to provide the cooperative with the best baseline data from these crops in these sites in order to affect a more titrated crop plan. The cooperative reported that these numbers proved highly accurate in the 2022 execution.

We believe that the microfarm model combined with cooperative aggregation and marketing offers genuine pathway for urban, suburban, and even rural specialty crop producers anywhere in the country, but especially in small cities across the upper Midwest. The Microfarm model – standard eases training, eases planning, scalable, also offer untapped promise for low cost automation and robotics. 6,000 – 8,000 has the promise to return $50,000-$70,000 a year on a high functioning, market connected (coop) site.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

We have used actual field and market data to make all of our estimates and conclusions. Our field data moves through three real world filters (transfer to the aggregation site, repackaging at aggregation site, point of sale) during which any errors are removed by necessity. Our evaluation and revision of the microfarm training program were based on real world results with our Mansfield farmers, extensive interviews with those farmers, professional peer review with Extension educators, and a second set of trainings performed in Marion, Ohio. Our conlusions abut the efficacy of the training is based on field results. Our assessment of the cooperative and farm businesses is based up on their actual market returns during and after the grant.

2.8. List key stakeholders that could be served by results of this project.

Land Grant Universities, Extension.
Community Farming and small farming groups,
Community organizations interested din food systems development,
Food desert neighborhoods,
Food insecurity programs,
Small colleges seeking to engage community in food systems

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sourceswill be reported under Information Products. (up to 1,000-word limit)

YEAR 2 Knowledge Dissemination:

PI Kent "Kip" Curtis- Named Faculty Fellow for the Office of Outreach and Engagement, through this support Dr. Curtis began a community outreach process in Marion, Ohio, through which he and community partner, Walter Bonham, creator of the Food Lab, LLC., shared knowledge in information about the pilot system in Mansfield via Marion web page and a series posted on YouTube.
• Presentation January 14, 2020 – Franklin County Local Food Policy Council
• "Local Food and the Food of Agriculture!" A Next World Conversations online discussion. Co-Panelist with Dr. Tim Van Meter, Associate Professor in the Alford Chair of Christian Education and Youth Ministry Coordinator of Ecological Initiatives, MTSO. Hosted by Dr. Anna Willow, Ohio State University. June 25, 2020
• "The Mansfield Microfarm Project: Agro-entrepreneurism as Economic Development in a Rust Belt City," as part of Panel B-5-3 Food and Agriculture: Environmental Justice at the Association for Environmental Studies and Sciences Annual Conference, July 16, 2020.Presentation for the Ohio State University Urban Extension Workgroup, August 14, 2020. Via zoom.
• "The Impacts of COVID-19 on Food Security and Long Term Implications and Adaptations," A Center for Urban and Regional Analysis Webinar, Panelist. September 18, 2020. https://youtu.be/rbNU2XVhvbc
Kareem Usher (OSU Faculty Researcher)

• Invited Guest Lecturer: University of Vermont – CDAE 004: US Food, Social Equity, and Development
(17th November, 2020).
• Invited Speaker: Graduate Professional Student Initiative, (26th October, 2020).
• Invited Panelist: 'Good, Clean, and Fair: Slow Food for All?' Spring 2020 Lecture Series. Hosted by the Ohio State University Anthropology Public Outreach Program and Slow Food Columbus, (22nd April, 2020).
• Invited Speaker: Loose Talk Series. Knowlton School of Architecture, (31st January, 2020).
• Invited Panelist: Center for Urban and Regional Analysis (The Ohio State University). Food Security
Panel Discussion (24th January, 2020) All Researchers, created on-line recording Virtual Urban Farming Symposium, October 26, 2020, via zoom

YEAR 3 Knowledge Dissemination:
Year 3 had multiple opportunities to discuss and disseminate both the whole microfarm system concept and some components of the research knowledge gained.
PI Kent "Kip" Curtis
Multiple public activities disseminating the knowledge gained from the FFAR implementation work:

- Presentation to the Annual Faculty Research "Frenzy," Mansfield Campus, 1/11/2021
- Development and recording of a TEDx Talk "Fulfilling the Promise of the Land Grant University", 1/27/2021
- Presentation for the Eminence Scholars Class of 2024, 4/21/2021
- Presentation to the Marion Minority Commission, 4/28/2021
- Presentation and tour for Senator Sherrod Brown, 5/4/2021
- Presentation to the Leadership Marion, 5/21/2021
- Presentation and tour to Dr. Kate Brown, MIT (writing a book about cooperatives), 6/4-5/2021
- Presentation to OSU Sustains Upward Bound program, 7/9/2021
- Presentation and tour to Mansfield Rotary Club, 9/7/2021
- Keynote Presentation at the Marion Microfarm Summit, 10/7/2021
- Presentation to the Hoover Foundation, 11/3/2021
- Presentation to the XPrize Racial Equity Initiative team, 11/29/2021
- Presentation to the farmOS development team, 12/8/2021
- Presentation to Shweta Narayan, business development capital, 12/14/2021

The presentations in Marion were part of a similar outreach process as the one in Mansfield and has already resulted in $200,000 in funding to help launch their initiative. Kate Brown's book will include a chapter about the Mansfield project and is due out in 2023. Presentation to Shweta Narayan has created a working relationship to begin exploring the possibilities for capitalizing needed technology to help improve the efficiency and profitability of the microfarm system.

Year 4 Knowledge Dissemination

- "Catalyzing Sustainable Change: Flipping the Script on Community Engagement," as part of "Food Systems as Engaged Opportunities," at the Engaged Scholars Consortium Annual Conference, September 22, 2022.
- "Cultivating Social Justice through a Food System Community of Practice," as part of "Engaged Communities: Opportunities and Success Stories" session at the Association for Environmental Studies and Sciences Annual Conference, Baltimore, MD, June 21, 2022.
- "Catalyzing Change in Marginalized Places," as part of "Environmental Justice: Building University-Community Collaborations" session at the American Society for Environmental History, Annual Meeting, Eugene, OR, March 24, 2022.
- "The Monsters You Don't See," as part of the "A Field of Monsters: What We Find at the Crossroads," The Mansfield Playhouse, Mansfield, Ohio, October 14, 2022.
- "Morality, Ethics, Race, and the Environment," presented at All Saints Church, Belleville, Ohio, September 18, 2022.
- "Microfarming as Economic Opportunity," presented to the Ohio State Legislature, Agricultural Committee, Columbus, Ohio, May 24, 2022.
- "Microfarming in Marion," at the Marion Spring Fling Gala Fundraiser, Marion, Ohio, May 20, 2022.
- "Catalyzing Sustainability with Microfarming," presented at the Environmental Justice Dinner Sponsored by the OSU Sustainability student group, OSU, Columbus, Ohio, April 6, 2022.

In addition to the multiple public and academic presentations given about the microfarm project, its success has attracted the attention of others interest in replicating the system. PI Kent "Kip" Curtis has consulted with the Mininster of Agriculture for St. Vincents and the Germains on the potential of microfarming in that small Caribbean nation. He is also working with the Barrytown Land Trust outside of Louisville, Kentucky where a residential Black community that was founded by Freedmen is under pressure from increasing land prices and is seeking to implement microfarming as a part of community development effort and to a partnership of food policy and food system constituents in Toledo, Ohio to explore similar plans. Finally, Antioch College in Yellow Spring, Ohio is exploring the possibility of converting its campus farm into a microfarming training center.

2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

COVID-19 fell in the middle year of project implementation. We were successful in shifting farmer training sessions and meetings online and quickly gaining permission from the university to follow-strict COVID restrictions and slowly assemble the remaining five microfarms during late summer and autumn. But it stopped the social science team in their tracks and they had great difficulty recovering.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Nothing to report.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reportingperiod. If other education components are involved, please describe them in detail. (up to 500-word limit)

Glennon Sweeney – Graduate Student in Urban Studies
Ives Hartman – Graduate Student in History
Frederick Michel graduate students (2)

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video,audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Information Products:

Kip Curtis, PI, became a member of the OSU University-wide Urban Extension interest group, which meets monthly to discuss urban extension initiatives and collaborations and has included the microfarm project in its development of knowledge resources for the web.

https://youtu.be/7yQQdbI2wK8
; https://youtu.be/ePRirVaCaRs
; https://youtu.be/KlGvLjp1EaM\

Created videotape explaining the design of the project in October 2020:
https://youtu.be/tWhGbhvZskc

Created a short documentary of the project and progress to date for our November 2020 virtual symposium:

https://youtu.be/WGQqTlCzaOU

During autumn 2020 I was interviewed for a story to run in 2021 in the College of Arts & Sciences news page.
https://artsandsciences.osu.edu/news/mansfield-microfarms-craft-sustainable-model-urban-agriculture

This is a list of the OSU Media Coverage throughout the life of the project:

Microfarm Project Media Coverage

Microfarm Project Establishing Roots in Marion Community with First Crops in the Ground
Marion Star, July 15, 2022

Receive Fresh, Locally Grow Produce Right to Your Door: Richalnd Gro-Op Announces New Subscription Box Program
Richland Source, June 27, 2022

Out of the Blank #1076 podcast – Kip Curtis
April 13, 2022

After Success in Mansfield, the Microfarm Project Launches in Marion
Marion Star, April 9, 2022

Mansfield School Board Approves High School Hoop House, Honors Prospect Secretary
Richland Source, March 9, 2022

All about Microfarms and The Ohio State Marion and Mansfield Campus Projects
That's Agra Tastic Show podcast, February 28, 2022

Marion City Schools to Offer Microfarming Career Pathway with $200,000 Grant
Marion Star, January 29, 2022

Leadership Marion to Host Summit on Urban Farming Opportunities
OSU Marion News, September 15, 2021

Microfarm Project Forges Ahead with New Funding and Expansion to Marion
Ohio State Discovery Themes, November 23, 2021

Urban farm project a growing success in Mansfield
Richland Source, Carl Hunnell, April 15, 2020

GALLERY: Urban Farms at Sixth and Bowman Streets in Mansfield
Knox Pages, Carl Hunnell, April 14, 2020

Ohio project partners urban, rural microfarms
University of Kentucky Center for Crop Diversification, Matt Ernst, Feb 2020

Growing Together: Mansfield microfarm cooperative heads into second year
Farm and Dairy, Sarah Donaldson, February 20, 2020

Ohio State Mansfield and FFAR Launch $2 Million Food System Project
FFAR Staff, May 2, 2019

Mansfield Microfarm Project will support economy, supply local produce
Ohio State News, May 2, 2019

Ohio State-Mansfield and food/ag researchers launch $2 million food system project
Richland Source, May 2, 2019

OSU Mansfield to pilot urban agriculture project
Farm and Dairy, May 2, 2019

'Micro-farm' to bring fresh produce to food deserts in Mansfield area
News Channel 5 Cleveland, May 9, 2019

'Micro-farm' to bring fresh produce to food deserts in Mansfield area
Yahoo News, WEWS Cleveland Videos, May 10, 2019

Brown Announces $2 Million for Ohio State University at Mansfield to Launch Sustainable Food System Project
Sherrod Brown Senator for Ohio, May 13, 2019

Ohio State University at Mansfield Launches $2M Sustainable Food System Project
Smart Cities Connect, May 15, 2019

Foundation for Food and Agriculture Research awards grant to Ohio State University to launch urban sustainable food system
Refrigerated and Frozen Foods, May 15, 2019

OSU-Mansfield ready to expand micro-farm program to Mansfield community
Mansfield News Journal, Al Lawrence, March 27, 2019

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WHAT IS MICRO FARMING?
Hello Homestead, Sam Schipani, July 10, 2019

Mansfield microfarm booms thanks to community support
Richland Source, Katie Hunt, July 25, 2019

OHIO STATE MANSFIELD STUDENT AND PROFESSOR CREATE URBAN AGRICULTURAL SYSTEM FOR FOOD DESERTS
The Lantern, Lily Maslia, September 5, 2019

Richland County Farm Bureau celebrates centennial
Richland county farm bureau, Eileen Eisenhauer, Oct 17, 2019

Mansfield Microfarm Project
The Working Landscapes Lab, 2019

Mansfield Microfarm Project will Support Economy, Supply Local Produce
Ohio State News, May 2, 2019

Symposium Generates Dialogue on Building a Better Food System in Richland Co
Richland Source, Emily Dech, March 1, 2018

OSU students are redefining urban agriculture
Morning Ag Clips, March 22, 2018

Micro-farm at Ohio State University-Mansfield could help feed residents
Mansfield Journal News, Tom Brennan, April 5, 2018

OSU Mansfield micro-farm a potential model for new food system
Richland Source, Tom Brennan, April 8, 2018

OSU-M microfarm project advances in Alliance for the American Dream competition
Richland Source, December 21, 2018

New 'Micro Farm' Model Tested At OSU Mansfield Urban Agriculture Project
Be Well podcast

Growing a Better Food System
Mansfield News Journal, April 28, 2017

Microfarm Growing at OSU Mansfield
1812 Blockhouse.com, September 13, 2017

OSU-M Seeks Land Bank's Help with Microfarms
Mansfield New Journal, September 16, 2017

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RUSS Garden Aims to Offer Hands-on Learning Opportunities to Local Students
Richland Source, October 19, 2017

Ohio State Mansfield builds micro-farm
Farm and Dairy, Catie Noyes, October 27, 2017

Ohio State Mansfield Debuts RUSS Gardens Project
WMFD, October 19, 2017

Mansfield micro-farm part of larger vision to impact local food system
Farm World, Doug Graves, November 17, 2017

When an Abandoned Parking Lot Becomes a Farm
OSU CFAES Newsletter, November 2017

Construction Begins for Micro-Farm on Ohio State Mansfield Campus
The Ohio State University-Mansfield

NECIC Media Coverage:
North End Community Improvement Collaborative Announces RICI Farming Program
The Richland Source, March 31, 2023

New 'Micro Farm' Model Tested At OSU Mansfield Urban Agriculture Project
Ideastream, Lucia Bushak, May 20, 2019

Producers begin harvesting crops at OSU-Mansfield microfarm
Mansfield News Journal, Monroe Trombly, May 21, 2019

$2 Million Food System Aims To Make Produce Accessible
WMFD, Jenna Ramolt

OSU Mansfield Micro-Farm Begins First Harvest
WMFD, Jesse Smith

Bonham: 6 new microfarms to take root in Mansfield
Mansfield News Journal, Tom Brennan, July 26, 2019

Urban farm planned for former Gorman-Rupp site on Mansfield's north end
Richland Source, July 28 2019

NECIC Urban Farm Opens With Ribbon Cutting
WMFD

Urban gardens planted to feed downtown Mansfield
Mansfield News Journal, Zach Tuggle, September 10, 2019

GrowFourth Media Coverage:
Ribbon-cutting officially launches the Stanfield family's 'GrowFourth Urban Farm'
Richland Source, Carl Hunnell, October 1, 2019

Stanfield family leads way for GrowFourth Urban Farm
Mansfield News Journal, Monroe Trombly, October 2, 2019

Fifth Graders Help at Urban Farm
St. Peter's School, Kathy Morris, October 19, 2019

3.2. Please provide a list of citations for the information products produced during theproject.

YEAR 2 Citations:

Adderley, B., Rodriguez, M. T., Sweeney, G., & Curtis, K., PI. Exploring the connection between household resilience and food security in a post-industrial city. Urban Food Systems Symposium, Kansas City, KS, United States. October 7, 2020.

Esha Shrestha, Ashish Manandhar, Fred Michel, Ajay Shah, and Kip Curtis, PI. Life cycle assessment of spinach grown in an urban farm in Mansfield, Ohio. Ohio State University, College of Food, Agriculture, and Environmental Science Annual Poster Session May 15, 2020.

YEAR 3 Citations:

PI Kent "Kip" Curtis
Conferences

• Curtis, K., West-Torrence, D., Bonham, W., Sweeney, G., "Catalyzing Sustainable Futures: Building Partnerships for Social Justice," under review with the Journal of Outreach and Engagement in Higher Education since January 2021. This article discusses the partnership building activities we engaged in Mansfield to develop the FFAR grant.

• Curtis, K., "Catalyzing Change in Marginalized Places: The Microfarm Project." This article locates the project within several bodies of literature (community based participatory research, communities of practice, Black farmer land loss, urban marginalization, urban renewal, urban sustainability, and urban agriculture) and details the implementation process of this work as a model effort for university-community partnerships. Expected submission in August, 2023.

Forbes Lipschitz
Conferences

• Lipschitz, F and Chris Strasbaugh "Multispectral Sensing for Evaluating the Landscape Impacts of Microfarms" 2021 Conference of the Council of Educators in Landscape Architecture, March 17, 2022, virtual due to Covid-19

Fred Michel
Publications

• Michel Jr., FC, E Shrestha, A Manandhar, A Shah. Is Local Food Production Better? Ohio Country Journal. August 2021.

• Michel Jr., FC, T O'Neill, R Rynk. Chapter 5-Passively aerated composting methods, including turned windrows. Ed: Robert Rynk, The Composting Handbook, Academic Press, 2022, Pages 159-196, ISBN 9780323856027, https://doi.org/10.1016/B978-0-323-85602-7.00002-9

• Michel Jr., FC, T O'Neill, R Rynk, J Gilbert, M Smith, J Aber, H Keener. 2022. Chapter 6 - Forced aeration composting, aerated static pile, and similar methods, Ed: Robert Rynk, The Composting Handbook, Academic Press, Pgs 197-269, ISBN 9780323856027, https://doi.org/10.1016/B978-0-323-85602-7.00007-8

• Michel Jr., FC, T O'Neill, R Rynk, M Bryant-Brown, V Calvez, Ji Li, J Paul. Chapter 7 - Contained and in-vessel composting methods and methods summary, Ed: Robert Rynk, The Composting Handbook, Academic Press, 2022, Pages 271-305, ISBN 9780323856027, https://doi.org/10.1016/B978-0-323-85602-7.00009-1.

Conferences

• Michel Jr. FC, K Curtis, G Sweeney, M Rodriguez, E Shrestha, A Shah. Sustainable Urban Agriculture in the US Post Industrial Midwest. International Society of Horticultural Science annual meeting. Abstract accepted for presentation at 2022 meeting.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during theproject resulting directly from the FFAR award? If yes, please provide citation.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

YEAR 2 Websites

Facebook: https://www.facebook.com/OhioStateMansfieldMicrofarm
This facebook page is maintained by the demonstration urban microfarm on the OSU Mansfield campus. It disseminates all news and information related to the whole project.

The Cooperative (RGO) Facebook Page: https://www.facebook.com/Richland-Gro-Op-298994960984805 This facebook page is maintained by the Richland Gro-Op and disseminates any news related to the cooperative and any of the microfarmers, highlights farmers and crops during the growing season, and serves as a key point of community outreach and engagement.

The Cooperative (RGO) Web Page: https://richlandgro-op.com/

YEAR 3 Websites

https://www.necic-ohio.org/programs/food-network/urban-farm
This website is maintained by community partner, NECIC to highlight its Urban Farm. It disseminates information related to the microfarms operating on that site, its Farmer's Market, and new and information related to Urban Farm activities.
https://richlandgro-op.com

This is the business web site for the Richland Gro-Op. It contains a description of the company history, profiles of its microfarmer operations, news and information about farmers and the cooperatives, and serves as the point of entry for direct sales in 2022.

https://osumarion.osu.edu/alumni-initiatives/initiatives/microfarm.html

This website is maintained by Ohio State University, Marion. It provides video and textual information related to the Mansfield project and the efforts to replicate this system in Marion, Ohio. It has also served as an outreach and engagement hub for registration information related to the Marion Urban Farming Summit and the upcoming Farmer Recruiting event.

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if theymust remain confidential indefinitely. (up to 500-word limit)

Yes. Crop and sales data are proprietary information owned by the Richland Gro-Op

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers,and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access andre-use these data in the future?

YEAR 2 Report

Starting in the summer of 2020, The Ohio State University Office of Outreach and Engagement supported Dr. Curtis through a faculty fellowship to allow him to focus a percentage of his time beginning a dialogue with nearby Marion, Ohio which also has an OSU regional campus and a small city demographic similar to Mansfield’s.

Additionally, the project hosted a virtual symposium in late 2020 during which two dozen participants were able to hear from our microfarm producers and cooperative managers as well as the core project team about the various lesson learned and challenges faced when trying to build a full production system from the ground up. In the process, Marion has produced additional informational materials and a website related to the expansion of the project: https://osumarion.osu.edu/alumni-initiatives/initiatives/microfarm.htm

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation ofexisting data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

None during this reporting period.

4.2. If you list multiple data sets, are these data sets related? If so, please provide ashort description of how they are related.

4.3. Please provide copies of relevant metadata records to support FFAR's mission ofenhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: _Kent Curtis___________________________________

PI Signature: ___ [signature] __ Date: ___June 6, 2023___

Authorized Signing Official (ASO) Name: Cheryl Sowash, Associate Director

ASO Signature: Cheryl Sowash Date: 6/6/2023